Supplementary Material: An Explainable Evolutionary Grammar Approach for Interpretable Data Stream Classification with Concept Drift

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This supplementary material provides extended experimental results, detailed statistical analyses, and per-dataset visualizations supporting the main paper. All sections are numbered with the prefix "S" to distinguish them from the main manuscript.

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1 Experimental Details

1.1 Dataset Dimensions

The experimental evaluation encompasses 48 datasets: 41 binary classification and 7 multiclass datasets. Table 1 shows the distribution across drift types.

Table 1: Dataset Distribution by Drift Type

Drift Type	**All**	**Binary Only**
Abrupt	16	14
Gradual	11	9
Noisy	8	8
Stationary	9	6
Real	4	4
Total	48	41

The 7 multiclass datasets are: LED_Abrupt_Simple, LED_Gradual_Simple, LED_Stationary, RBF_Stationary, WAVEFORM_Abrupt_Simple, WAVEFORM_Gradual_Simple, and WAVEFORM_Stationary. IntelLabSensors was excluded from all analyses due to data quality issues.

1.2 Experimental Configurations

Table 2 summarizes the seven EGIS configurations evaluated. All configurations use a stream of 12,000 instances divided into chunks of the specified size.

Table 2: EGIS Experimental Configurations

Label	**Chunk Size**	**Penalty (γ)**	**Num Chunks**	**Total Instances**
EXP-1000-NP	1000	0.0	12	12,000
EXP-1000-P	1000	0.1	12	12,000
EXP-2000-NP	2000	0.0	6	12,000
EXP-2000-P	2000	0.1	6	12,000
EXP-500-NP	500	0.0	24	12,000
EXP-500-P	500	0.1	24	12,000
EXP-500-P03	500	0.3	24	12,000

1.3 EGIS Hyperparameters

Table 3: EGIS Hyperparameter Settings

Parameter	**Value**
Population size	120
Max generations	200
Elitism rate	0.1
Intelligent mutation rate	0.8
Tournament size (initial/final)	2 / 5
Crossover rate	0.9
Hill climbing iterations	50
Gene Therapy injection rate	0.3
Drift detection method	ADWIN
Memory management	Adaptive sliding window

2 Complete Performance Tables

This section provides the complete per-dataset G-Mean performance tables for all configurations and models. These tables extend the summary statistics presented in the main paper.

2.1 Per-Dataset G-Mean: Binary Datasets, Chunk Size 500

Table 4: Performance Comparison on Binary Datasets (G-Mean) - Chunk Size 500

Dataset	EGIS	ARF	HAT	SRP	ROSE	eRulesD2S	CDCMS
Abrupt Drift (14)							
AGRAWAL_Abrupt_Chain_Long	**0.888**	0.788	0.664	0.785	0.830	0.528	0.652
AGRAWAL_Abrupt_Simple_Mild	**0.906**	0.825	0.585	0.866	0.900	0.542	0.622
AGRAWAL_Abrupt_Simple_Severe	0.900	0.840	0.680	0.889	**0.925**	0.543	0.690
HYPERPLANE_Abrupt_Simple	0.756	0.835	0.882	0.771	0.863	0.525	**0.900**
RANDOMTREE_Abrupt_Recurring	0.615	0.533	0.533	0.524	**0.654**	0.552	0.534
RANDOMTREE_Abrupt_Simple	0.614	0.550	0.542	0.511	**0.659**	0.542	0.524
RBF_Abrupt_Blip	0.832	0.910	0.837	**0.916**	0.896	0.521	0.851
RBF_Abrupt_Severe	0.807	0.894	0.789	**0.901**	0.870	0.522	0.791
SEA_Abrupt_Chain	0.947	0.972	0.930	0.956	**0.977**	0.623	0.935
SEA_Abrupt_Recurring	0.945	**0.977**	0.946	0.939	0.975	0.646	0.952
SEA_Abrupt_Simple	0.950	0.977	0.952	0.973	**0.979**	0.620	0.940
SINE_Abrupt_Simple	0.951	**0.988**	0.971	0.987	0.980	0.655	0.973
STAGGER_Abrupt_Chain	0.968	**0.994**	0.763	0.993	0.922	0.778	0.982
STAGGER_Abrupt_Recurring	0.957	0.997	0.790	0.989	**0.997**	0.797	0.988
Gradual Drift (9)							
HYPERPLANE_Gradual_Simple	0.756	0.832	0.879	0.774	0.863	0.536	**0.900**
RANDOMTREE_Gradual_Simple	0.610	0.549	0.523	0.511	**0.669**	0.561	0.522
RBF_Gradual_Moderate	0.819	0.895	0.820	**0.904**	0.889	0.535	0.820
RBF_Gradual_Severe	0.823	**0.902**	0.830	0.897	0.894	0.526	0.817
SEA_Gradual_Recurring	0.939	**0.961**	0.920	0.954	0.960	0.635	0.920
SEA_Gradual_Simple_Fast	0.953	0.977	0.950	0.976	**0.980**	0.638	0.951
SEA_Gradual_Simple_Slow	0.957	0.978	0.948	0.976	**0.981**	0.633	0.933
SINE_Gradual_Recurring	0.917	**0.952**	0.928	0.950	0.943	0.636	0.923
STAGGER_Gradual_Chain	0.920	**0.950**	0.686	0.936	0.920	0.764	0.920
Noisy Drift (8)							
AGRAWAL_Abrupt_Simple_Severe_Noise	0.917	0.779	0.687	0.836	**0.940**	0.547	0.658
HYPERPLANE_Gradual_Noise	0.753	0.811	0.856	0.759	0.848	0.530	**0.870**
RANDOMTREE_Gradual_Noise	0.620	0.511	0.532	0.477	**0.690**	0.550	0.516
RBF_Abrupt_Blip_Noise	0.827	0.904	0.825	**0.905**	0.886	0.523	0.846
RBF_Gradual_Severe_Noise	0.824	0.890	0.818	**0.901**	0.877	0.518	0.820
SEA_Abrupt_Chain_Noise	0.946	0.966	0.917	0.937	**0.971**	0.632	0.926
SINE_Abrupt_Recurring_Noise	0.913	0.981	0.949	**0.981**	0.973	0.668	0.945
STAGGER_Abrupt_Chain_Noise	0.965	**0.995**	0.758	0.982	0.922	0.774	0.942
Stationary Drift (7)							
AGRAWAL_Stationary	**1.000**	0.934	0.981	0.998	0.997	0.524	0.977
HYPERPLANE_Stationary	0.723	0.819	0.875	0.762	0.852	0.526	**0.902**
RANDOMTREE_Stationary	0.542	0.585	0.432	0.555	**0.594**	0.528	0.515
RBF_Stationary	0.814	0.921	0.900	**0.969**	0.963	0.268	–
SEA_Stationary	0.954	0.980	0.950	0.962	**0.984**	0.612	0.945
SINE_Stationary	0.953	0.986	0.968	**0.987**	0.982	0.635	0.976
STAGGER_Stationary	**1.000**	0.998	0.801	0.997	0.999	0.787	0.999
Real Drift (4)							
AssetNegotiation_F2	0.949	0.967	0.947	0.972	**0.982**	0.651	0.948
AssetNegotiation_F3	**0.995**	0.966	0.961	0.921	0.992	0.762	0.944
AssetNegotiation_F4	0.906	0.946	0.939	0.936	0.662	0.562	**0.948**
Electricity	0.710	0.887	0.861	0.885	**0.905**	0.619	0.857
EGIS W/L/D	–	11/31/0	26/16/0	13/29/0	8/34/0	42/0/0	24/17/0
Mean	0.858	0.879	0.817	0.871	0.894	0.597	0.843
Std	0.123	0.139	0.149	0.149	0.110	0.100	0.152
Avg Rank	4.02	2.64	4.88	3.26	2.26	6.50	4.37

2.2 Per-Dataset G-Mean: Binary Datasets, Chunk Size 1000

Table 5: Performance Comparison on Binary Datasets (G-Mean) - Chunk Size 1000

Dataset	EGIS	ARF	HAT	SRP	ROSE	eRulesD2S	CDCMS
Abrupt Drift (14)							
AGRAWAL_Abrupt_Chain_Long	**0.872**	0.790	0.676	0.861	0.830	0.526	0.622
AGRAWAL_Abrupt_Simple_Mild	**0.912**	0.799	0.646	0.803	0.900	0.537	0.607
AGRAWAL_Abrupt_Simple_Severe	0.902	0.847	0.690	0.857	**0.925**	0.537	0.684
HYPERPLANE_Abrupt_Simple	0.781	0.831	0.877	0.773	0.863	0.524	**0.896**
RANDOMTREE_Abrupt_Recurring	0.645	0.561	0.520	0.530	**0.654**	0.548	0.509
RANDOMTREE_Abrupt_Simple	0.649	0.548	0.492	0.501	**0.659**	0.550	0.524
RBF_Abrupt_Blip	0.858	0.914	0.836	**0.915**	0.896	0.527	0.837
RBF_Abrupt_Severe	0.832	**0.898**	0.812	0.896	0.870	0.508	0.777
SEA_Abrupt_Chain	0.962	0.972	0.930	0.931	**0.977**	0.618	0.933
SEA_Abrupt_Recurring	0.966	**0.977**	0.945	0.931	0.975	0.644	0.954
SEA_Abrupt_Simple	0.964	0.975	0.949	0.961	**0.979**	0.617	0.940
SINE_Abrupt_Simple	0.964	**0.987**	0.975	0.986	0.980	0.650	0.969
STAGGER_Abrupt_Chain	0.931	**0.995**	0.781	0.992	0.922	0.756	0.970
STAGGER_Abrupt_Recurring	0.909	**0.998**	0.786	0.991	0.997	0.763	0.978
Gradual Drift (9)							
HYPERPLANE_Gradual_Simple	0.781	0.837	0.879	0.771	0.863	0.540	**0.896**
RANDOMTREE_Gradual_Simple	0.654	0.513	0.527	0.487	**0.669**	0.537	0.542
RBF_Gradual_Moderate	0.847	0.896	0.819	**0.902**	0.889	0.514	0.823
RBF_Gradual_Severe	0.848	**0.905**	0.825	**0.905**	0.894	0.520	0.840
SEA_Gradual_Recurring	0.942	0.959	0.914	0.933	**0.960**	0.648	0.906
SEA_Gradual_Simple_Fast	0.965	0.978	0.947	0.932	**0.980**	0.600	0.945
SEA_Gradual_Simple_Slow	0.962	0.977	0.945	0.972	**0.981**	0.629	0.934
SINE_Gradual_Recurring	0.923	**0.952**	0.936	0.950	0.943	0.646	0.918
STAGGER_Gradual_Chain	0.878	**0.949**	0.722	0.943	0.920	0.780	0.911
Noisy Drift (8)							
AGRAWAL_Abrupt_Simple_Severe_Noise	0.928	0.809	0.660	0.749	**0.940**	0.545	0.630
HYPERPLANE_Gradual_Noise	0.758	0.822	0.855	0.763	0.848	0.537	**0.873**
RANDOMTREE_Gradual_Noise	0.651	0.532	0.552	0.464	**0.690**	0.536	0.535
RBF_Abrupt_Blip_Noise	0.853	0.902	0.824	**0.904**	0.886	0.519	0.829
RBF_Gradual_Severe_Noise	0.842	0.896	0.816	**0.900**	0.877	0.526	0.815
SEA_Abrupt_Chain_Noise	0.957	0.968	0.929	0.936	**0.971**	0.597	0.923
SINE_Abrupt_Recurring_Noise	0.884	0.980	0.953	**0.981**	0.973	0.627	0.927
STAGGER_Abrupt_Chain_Noise	0.933	**0.995**	0.764	0.987	0.922	0.777	0.925
Stationary Drift (7)							
AGRAWAL_Stationary	**1.000**	0.941	0.980	0.998	0.997	0.531	0.979
HYPERPLANE_Stationary	0.737	0.820	0.881	0.762	0.852	0.525	**0.908**
RANDOMTREE_Stationary	0.561	0.584	0.496	0.540	**0.594**	0.514	0.501
RBF_Stationary	0.835	0.927	0.900	**0.974**	0.963	0.283	–
SEA_Stationary	0.971	0.979	0.958	0.950	**0.984**	0.625	0.948
SINE_Stationary	0.966	0.987	0.971	**0.988**	0.982	0.635	0.975
STAGGER_Stationary	**1.000**	0.997	0.805	0.997	0.999	0.761	0.999
Real Drift (4)							
AssetNegotiation_F2	0.964	0.958	0.951	0.973	**0.982**	0.627	0.936
AssetNegotiation_F3	**0.998**	0.957	0.967	0.929	0.992	0.761	0.954
AssetNegotiation_F4	0.915	0.947	0.939	0.936	0.662	0.572	**0.947**
Electricity	0.774	0.891	0.856	0.854	**0.905**	0.638	0.830
EGIS W/L/D	–	12/30/0	31/11/0	21/21/0	8/34/0	42/0/0	29/12/0
Mean	0.868	0.880	0.821	0.864	0.894	0.592	0.838
Std	0.112	0.138	0.145	0.152	0.110	0.096	0.153
Avg Rank	3.57	2.69	4.76	3.48	2.26	6.57	4.61

2.3 Summary Statistics by Configuration and Model

Table 6: Mean G-Mean by Configuration and Model

Config	Model	Mean G-Mean	Std	N
EXP-1000-NP	ACDWM	0.8182	0.0781	41
EXP-1000-NP	ARF	0.8726	0.1309	48
EXP-1000-NP	CDCMS	0.8414	0.1520	41
EXP-1000-NP	EGIS	0.8657	0.1137	48
EXP-1000-NP	ERulesD2S	0.5558	0.1358	48
EXP-1000-NP	HAT	0.7698	0.2297	48
EXP-1000-NP	ROSE	0.8551	0.1702	48
EXP-1000-NP	SRP	0.8478	0.1521	48
EXP-1000-P	EGIS	0.8663	0.1136	48
EXP-2000-NP	ACDWM	0.7399	0.0686	41
EXP-2000-NP	ARF	0.8695	0.1333	48
EXP-2000-NP	CDCMS	0.8385	0.1534	41
EXP-2000-NP	EGIS	0.8551	0.1097	48
EXP-2000-NP	ERulesD2S	0.5555	0.1335	48
EXP-2000-NP	HAT	0.7721	0.2289	48
EXP-2000-NP	ROSE	0.8551	0.1702	48
EXP-2000-NP	SRP	0.8499	0.1504	48
EXP-2000-P	EGIS	0.8472	0.1139	48
EXP-500-NP	ACDWM	0.8603	0.0929	41
EXP-500-NP	ARF	0.8624	0.1402	48
EXP-500-NP	CDCMS	0.8460	0.1518	41
EXP-500-NP	EGIS	0.8561	0.1238	48
EXP-500-NP	ERulesD2S	0.5616	0.1371	48
EXP-500-NP	HAT	0.7667	0.2310	48
EXP-500-NP	ROSE	0.8551	0.1702	48
EXP-500-NP	SRP	0.8464	0.1648	48
EXP-500-P	EGIS	0.8549	0.1224	48
EXP-500-P03	EGIS	0.8365	0.1384	48

3 Extended Statistical Analysis

The main paper presents statistical analysis on the 41 binary datasets as the primary comparison. This section provides the complementary all-48 dataset analysis and detailed pairwise comparisons across all configurations.

3.1 All-48 Datasets Friedman Ranking

On all 48 datasets, the Friedman test yields $\chi^2(7) = 168.0$ ($p < 0.001$) with Nemenyi critical distance CD = 1.52.

3.2 Pairwise Wilcoxon Tests Across Configurations

Table 8 summarizes pairwise Wilcoxon tests (EGIS vs each baseline) across all configurations, including raw p-values, Bonferroni-adjusted p-values, and Cliff's delta effect sizes.

Table 7: Model Ranking on All 48 Datasets (EXP-2000-NP)

Rank	Model	Avg Rank	Mean G-Mean
1	ROSE	2.42	0.855
2	ARF	2.88	0.870
3	SRP	3.44	0.850
4	**EGIS**	3.62	0.855
5	CDCMS	4.92	0.716
6	HAT	4.98	0.772
7	ACDWM	6.44	0.632
8	ERulesD2S	7.31	0.556

Friedman: $\chi^2(7) = 168.0$, *p ¡ 0.001, CD = 1.52*

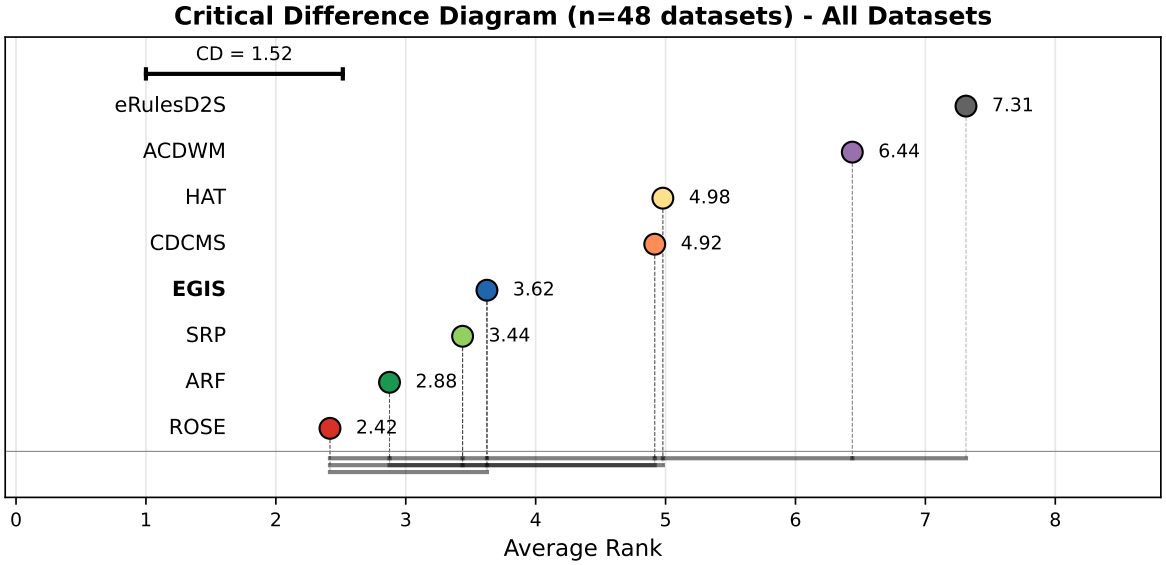

Figure 1: Critical difference diagram for all 48 datasets (EXP-2000-NP). Methods connected by a horizontal bar are not significantly different (Nemenyi, $\alpha = 0.05$).

Table 8: Pairwise Wilcoxon Tests: EGIS vs Baselines (Binary Datasets)

Config	Comparison	Raw p	Adj. p	Sig.	δ	Effect
EXP-1000-NP	ARF	0.1633	1.0000	No	-0.133	negligible
EXP-1000-NP	SRP	0.8778	1.0000	No	-0.055	negligible
EXP-1000-NP	HAT	0.0007	0.0052	Yes	0.197	small
EXP-1000-NP	ROSE	<0.0001	<0.0001	Yes	-0.179	small
EXP-1000-NP	ACDWM	<0.0001	<0.0001	Yes	0.402	medium
EXP-1000-NP	ERulesD2S	<0.0001	<0.0001	Yes	0.926	large
EXP-1000-NP	CDCMS	0.0929	0.6505	No	0.084	negligible
EXP-2000-NP	ARF	0.0437	0.3062	No	-0.200	small
EXP-2000-NP	SRP	0.5811	1.0000	No	-0.135	negligible
EXP-2000-NP	HAT	0.0061	0.0428	Yes	0.115	negligible
EXP-2000-NP	ROSE	<0.0001	0.0002	Yes	-0.248	small
EXP-2000-NP	ACDWM	<0.0001	<0.0001	Yes	0.611	large
EXP-2000-NP	ERulesD2S	<0.0001	<0.0001	Yes	0.930	large
EXP-2000-NP	CDCMS	0.2247	1.0000	No	0.002	negligible
EXP-500-NP	ARF	0.0757	0.5298	No	-0.172	small
EXP-500-NP	SRP	0.0825	0.5773	No	-0.116	negligible
EXP-500-NP	HAT	0.0274	0.1921	No	0.180	small
EXP-500-NP	ROSE	<0.0001	0.0006	Yes	-0.184	small
EXP-500-NP	ACDWM	0.9535	1.0000	No	0.133	negligible
EXP-500-NP	ERulesD2S	<0.0001	<0.0001	Yes	0.843	large
EXP-500-NP	CDCMS	0.5908	1.0000	No	0.030	negligible

4 Extended Transition Metrics

This section provides detailed transition metric analysis across all configurations, extending the EXP-2000-NP results presented in the main paper.

4.1 Transition Metrics by Configuration

Table 9: Transition Metrics by Configuration and Drift Type

Config	Drift Type	TCS	RIR	AMS
EXP-1000-NP	Abrupt	0.236±0.113	0.252±0.202	0.285±0.091
EXP-1000-NP	Gradual	0.213±0.095	0.211±0.161	0.273±0.082
EXP-1000-NP	Noisy	0.226±0.092	0.218±0.167	0.301±0.063
EXP-1000-NP	Stationary	0.217±0.129	0.224±0.222	0.270±0.097
EXP-1000-NP	Real	0.221±0.092	0.226±0.160	0.265±0.062
EXP-1000-P	Abrupt	0.236±0.112	0.251±0.204	0.284±0.091
EXP-1000-P	Gradual	0.217±0.097	0.218±0.164	0.275±0.083
EXP-1000-P	Noisy	0.227±0.090	0.220±0.161	0.300±0.062
EXP-1000-P	Stationary	0.219±0.131	0.231±0.226	0.268±0.095
EXP-1000-P	Real	0.214±0.092	0.211±0.162	0.268±0.062
EXP-2000-NP	Abrupt	0.284±0.128	0.346±0.229	0.266±0.101
EXP-2000-NP	Gradual	0.233±0.128	0.243±0.228	0.275±0.073
EXP-2000-NP	Noisy	0.265±0.130	0.301±0.255	0.267±0.093

Config	Drift Type	TCS	RIR	AMS
EXP-2000-NP	Stationary	0.171±0.096	0.147±0.144	0.255±0.099
EXP-2000-NP	Real	0.193±0.092	0.187±0.157	0.243±0.048
EXP-2000-P	Abrupt	0.290±0.128	0.357±0.220	0.276±0.106
EXP-2000-P	Gradual	0.268±0.131	0.311±0.228	0.265±0.095
EXP-2000-P	Noisy	0.320±0.122	0.391±0.200	0.287±0.065
EXP-2000-P	Stationary	0.284±0.126	0.352±0.214	0.257±0.091
EXP-2000-P	Real	0.247±0.091	0.251±0.149	0.281±0.047
EXP-500-NP	Abrupt	0.234±0.095	0.245±0.169	0.287±0.094
EXP-500-NP	Gradual	0.212±0.069	0.200±0.125	0.286±0.081
EXP-500-NP	Noisy	0.229±0.084	0.223±0.154	0.303±0.066
EXP-500-NP	Stationary	0.213±0.094	0.213±0.159	0.269±0.093
EXP-500-NP	Real	0.215±0.087	0.220±0.160	0.261±0.058
EXP-500-P	Abrupt	0.230±0.100	0.239±0.166	0.288±0.097
EXP-500-P	Gradual	0.212±0.081	0.201±0.135	0.293±0.088
EXP-500-P	Noisy	0.229±0.077	0.219±0.145	0.304±0.061
EXP-500-P	Stationary	0.218±0.109	0.229±0.181	0.260±0.107
EXP-500-P	Real	0.209±0.070	0.203±0.122	0.265±0.057
EXP-500-P03	Abrupt	0.276±0.134	0.327±0.235	0.282±0.099
EXP-500-P03	Gradual	0.265±0.130	0.301±0.234	0.286±0.082
EXP-500-P03	Noisy	0.282±0.111	0.323±0.213	0.304±0.068
EXP-500-P03	Stationary	0.267±0.137	0.321±0.236	0.263±0.102
EXP-500-P03	Real	0.220±0.082	0.222±0.150	0.267±0.061

4.2 TCS Time Series

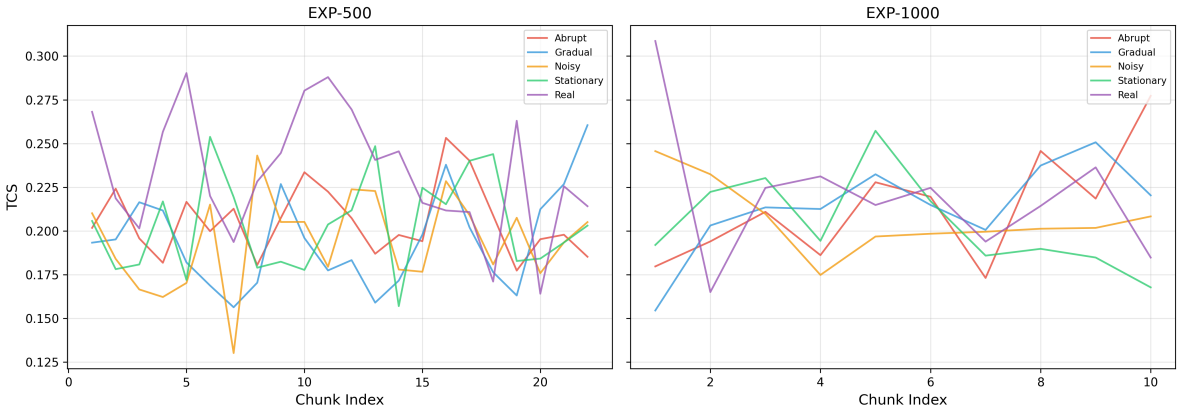

Figure 2: Transition Change Score (TCS) time series by drift type. Abrupt drift scenarios show characteristic spikes at drift points; gradual drift exhibits smoother transitions; stationary streams maintain consistently low TCS values.

4.3 TCS Comparison Across Configurations

5 EGIS Configuration Analysis

This section provides detailed analysis of how EGIS performance varies across chunk size and penalty configurations.

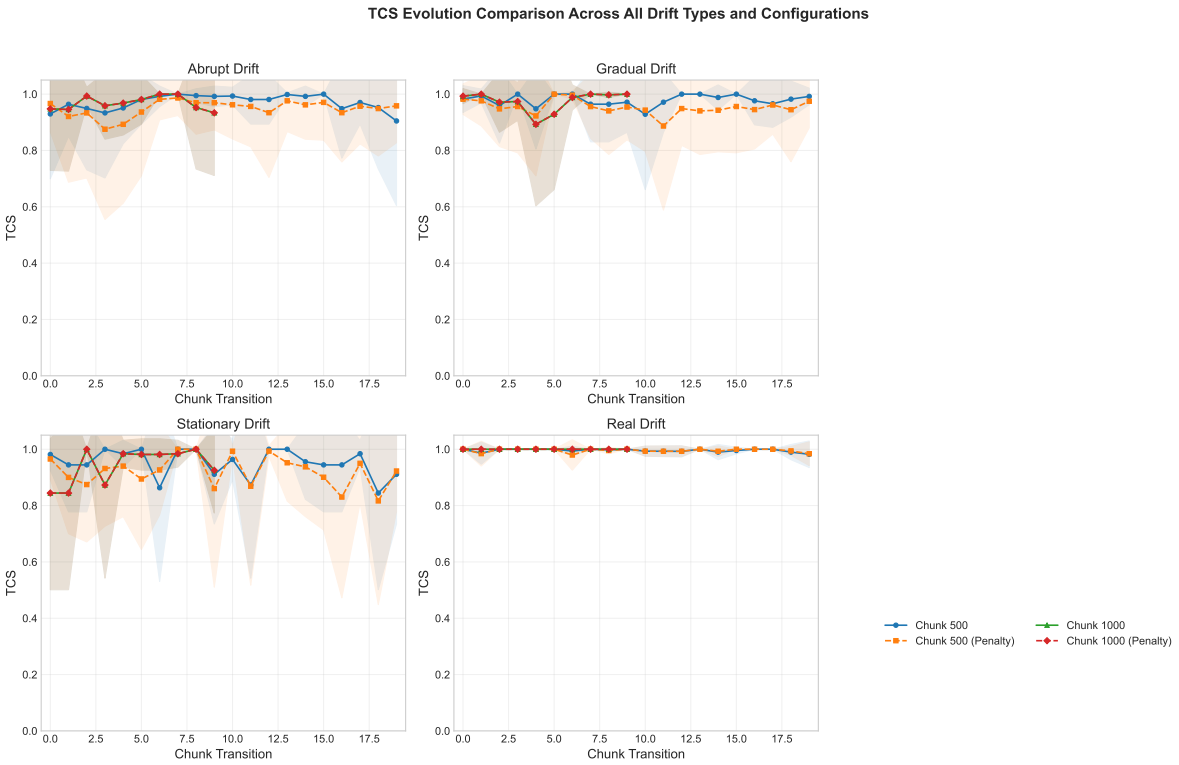

Figure 3: TCS comparison across drift types and configurations. Box plots show distributions of transition scores for each scenario.

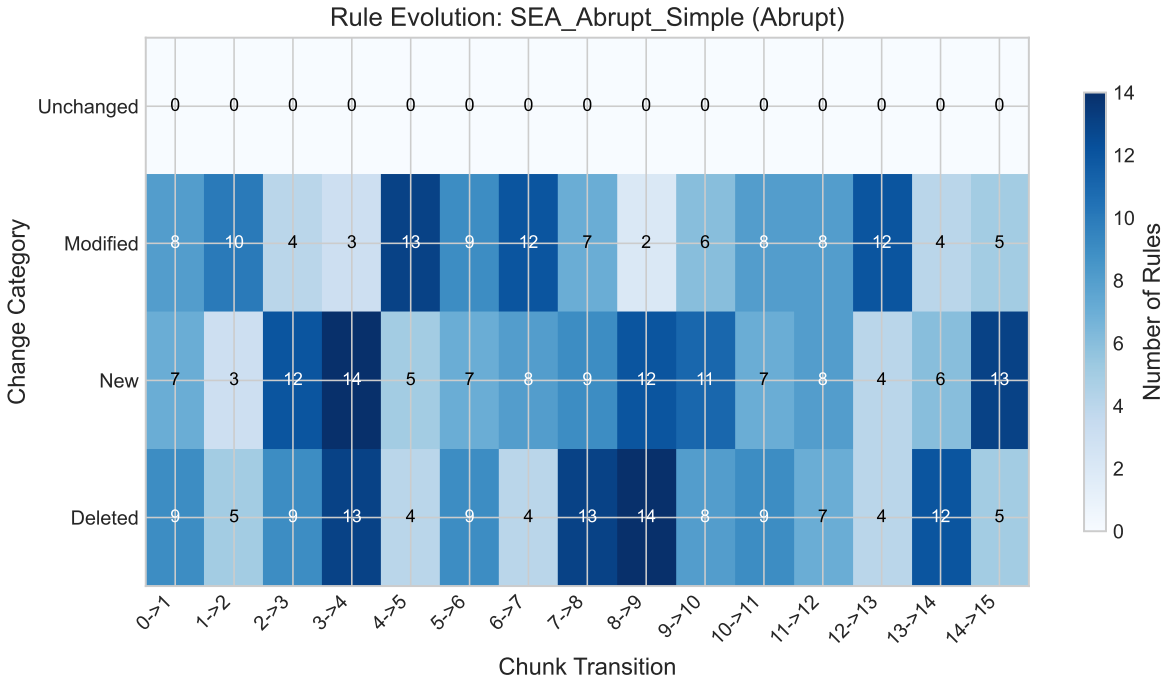

Figure 4: Rule evolution heatmap for abrupt drift, showing feature importance changes over time.

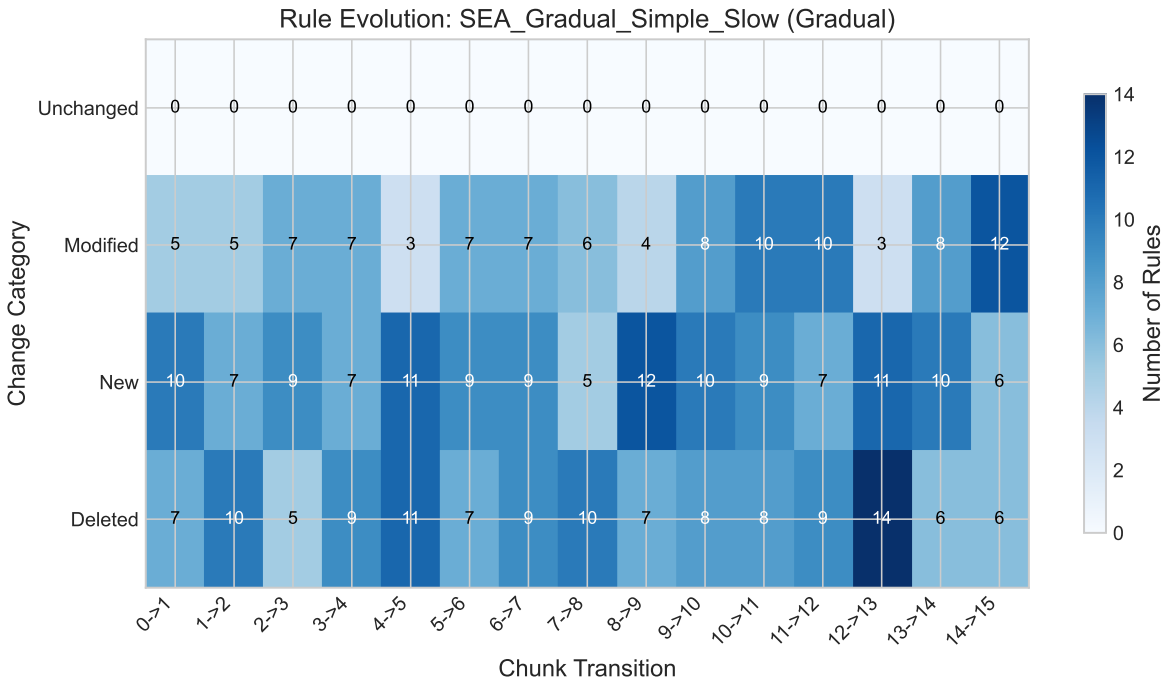

Figure 5: Rule evolution heatmap for gradual drift, showing feature importance changes over time.

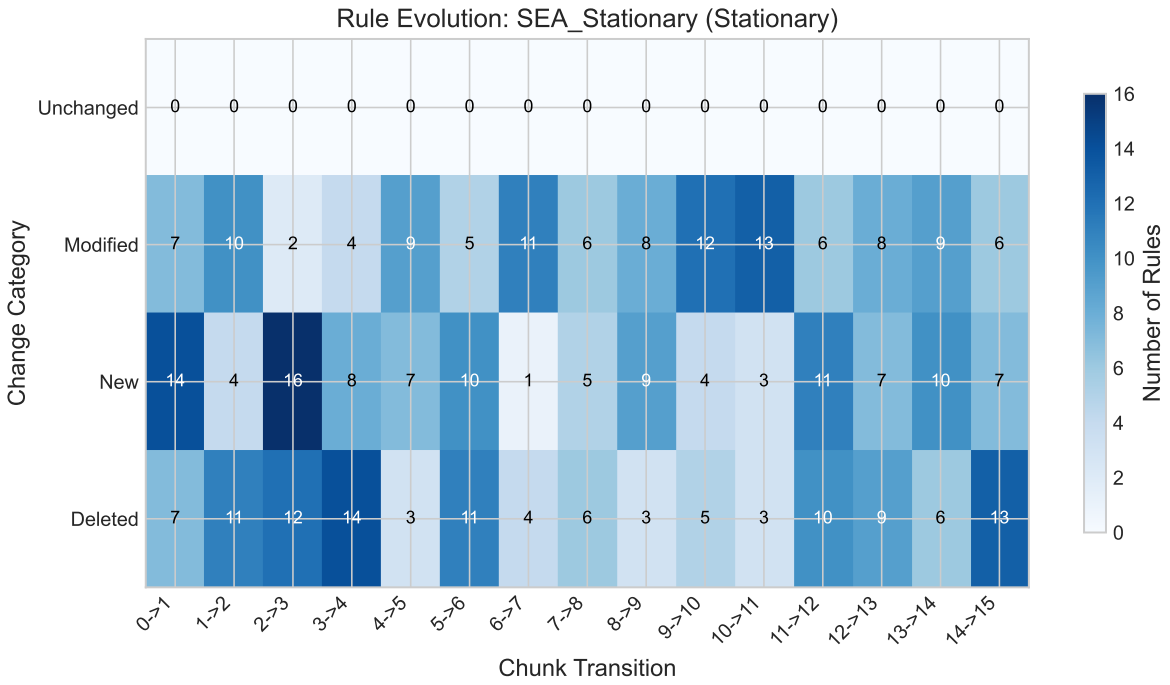

Figure 6: Rule evolution heatmap for stationary drift, showing feature importance changes over time.

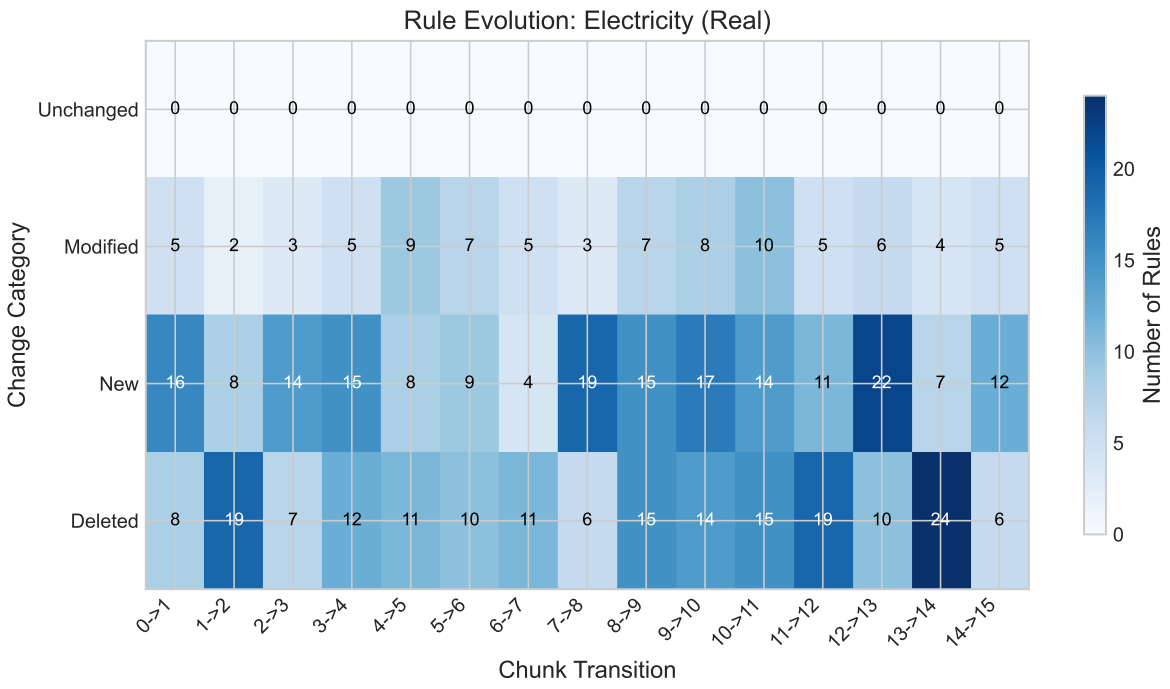

Figure 7: Rule evolution heatmap for real-world drift, showing feature importance changes over time.

5.1 Chunk Size Effect

5.2 Configuration Comparison

5.3 Performance Heatmap by Drift Type

5.4 Performance by Drift Type

6 Per-Dataset Visual Analysis

Per-dataset plots are available in the online supplementary archive. Each dataset has four visualizations: rule evolution matrix, attribute usage, rule components heatmap, and accuracy with detected drifts.

7 Rule Evolution Details

This section provides summary statistics on rule evolution patterns across configurations.

Table 10: Rule Evolution Summary by Configuration (Mean Counts per Transition)

Config	Unchanged	Modified	New	Deleted
EXP-1000-NP	0.2	9.4	10.1	10.0
EXP-1000-P	0.2	9.3	10.1	9.8
EXP-2000-NP	0.2	12.6	10.8	10.9
EXP-2000-P	0.3	6.5	9.7	10.4
EXP-500-NP	0.2	6.5	8.3	8.2
EXP-500-P	0.4	5.5	7.5	7.5
EXP-500-P03	0.3	4.6	7.1	7.2

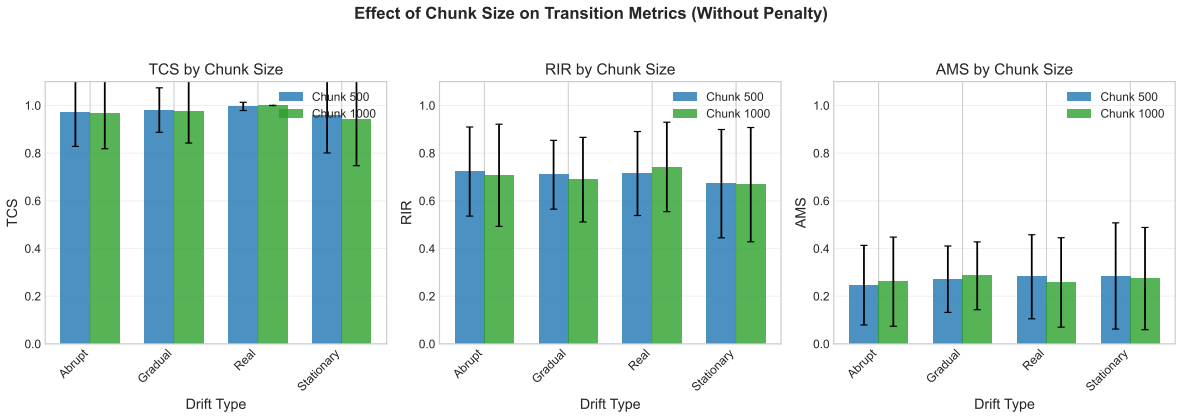

Figure 8: Chunk size sensitivity analysis showing EGIS performance (G-Mean) across configurations. Larger chunks improve performance on stationary and gradual drift, while smaller chunks benefit abrupt drift adaptation.

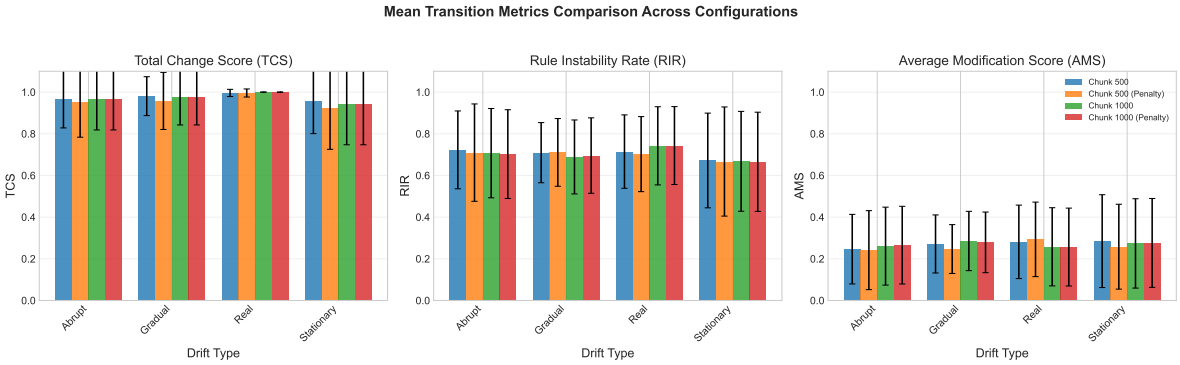

Figure 9: Configuration comparison showing the combined effect of chunk size and complexity penalty on EGIS performance. Error bars indicate standard deviation across datasets.

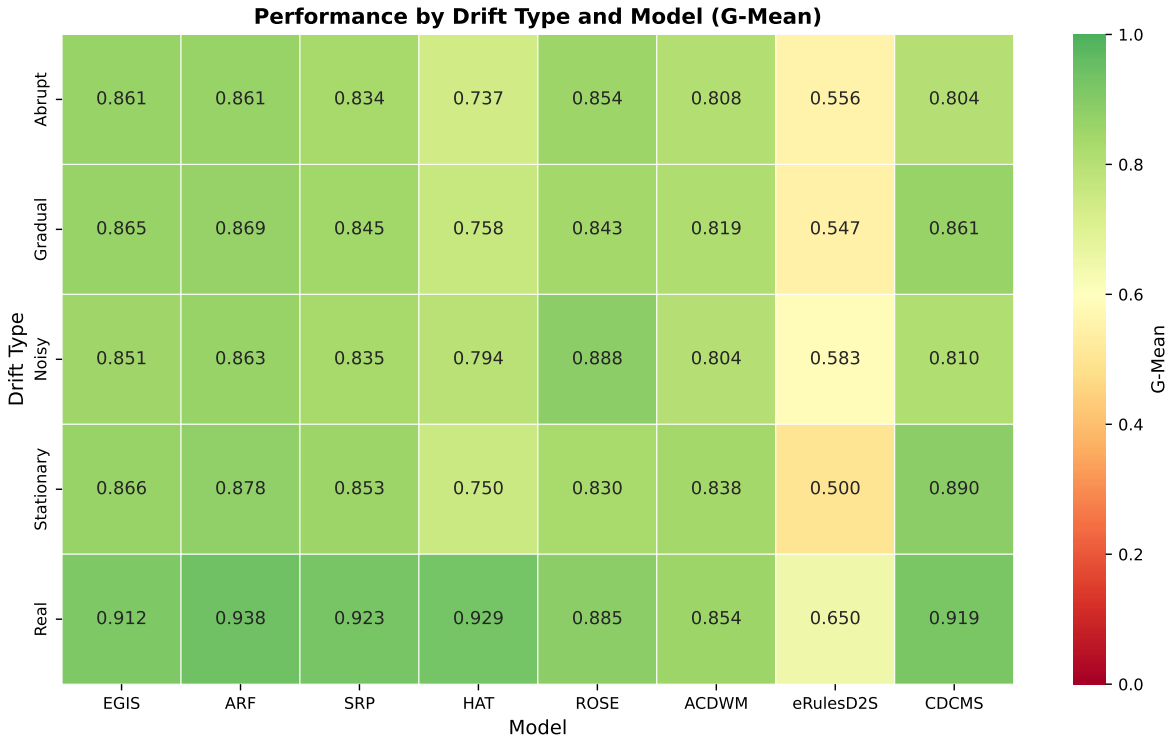

Figure 10: Performance heatmap by drift type and model (G-Mean) for binary datasets. Darker colors indicate higher performance.

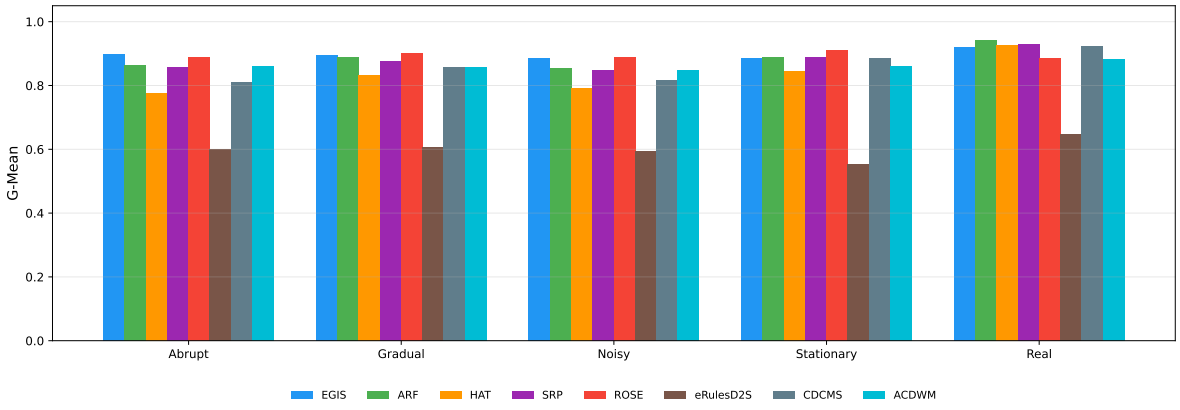

Figure 11: Performance comparison by drift type showing G-Mean for each model across drift categories.